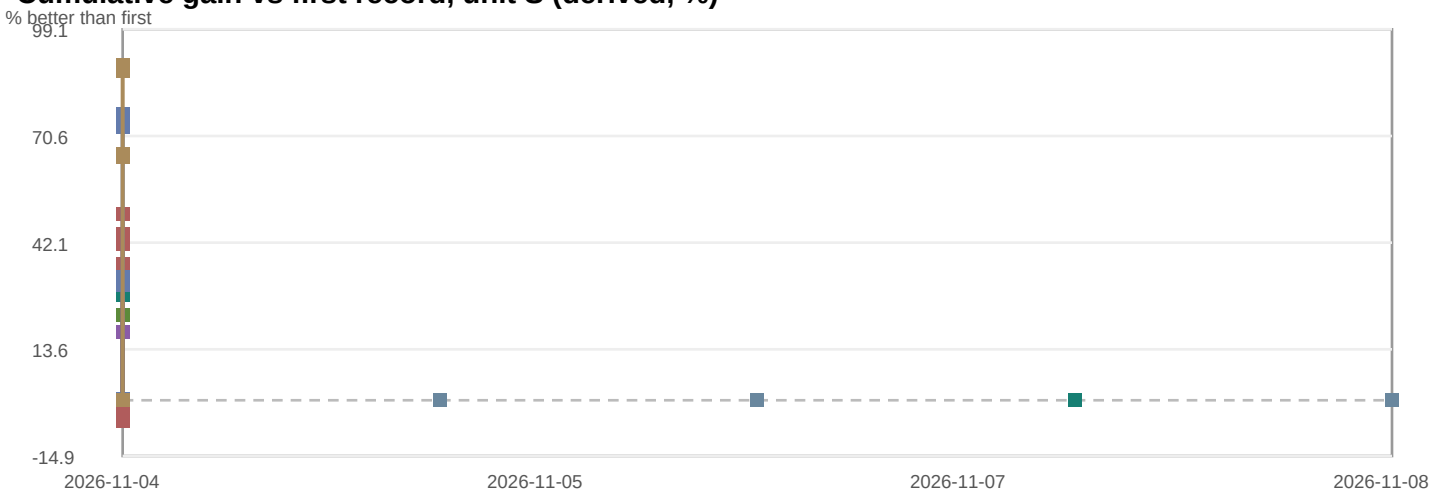
# Benchmark progress: crypto

From bench.record/v1 records in Git; raw data in S3 by SHA-256. Derived numbers are labelled derived.

Records: open 5, measured 56, invalid 0, censored 3. Headline unit S: group operations / sqrt(n); the ECDLP cost unit, every phase priced.

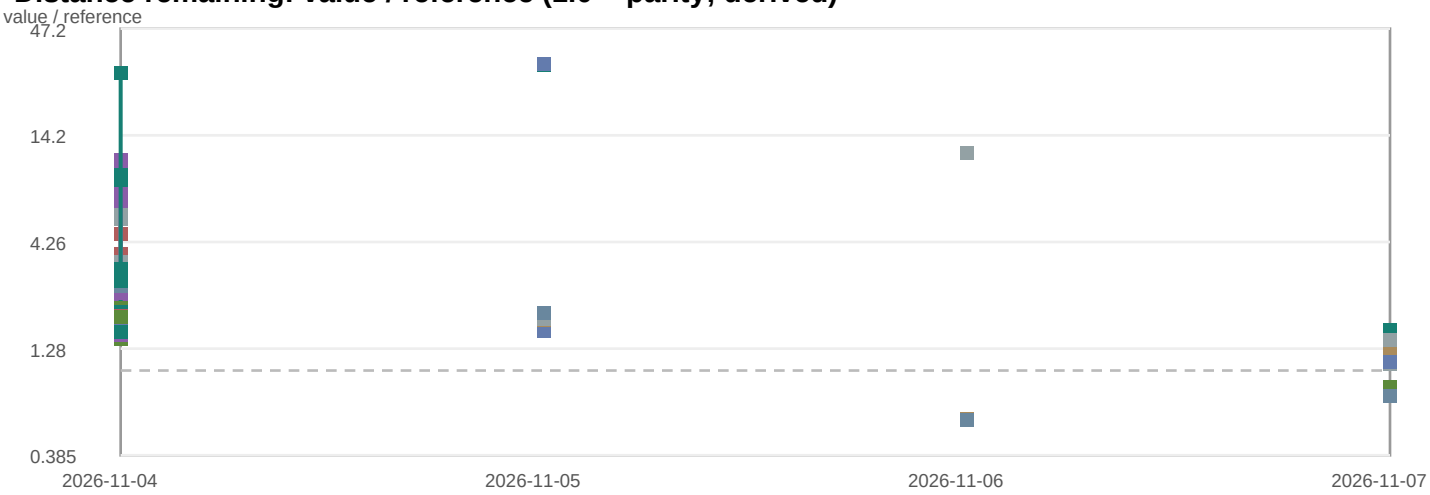
## Cumulative gain vs first record, unit S (derived, %)




### Series key

- archive:ic-packed-filter-p32-2
- archive:ic-packed-filter-p32-2
- dlpwai:static-dh-p224-shape •
- dlpwai:static-dh-p224-shape •
- dlpwai:static-dh-p224-shape •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20 • s_w
- ic:curve25519-family-k24-28 •
- ic:curve25519-family-k24-28 •
- ic:curve25519-family-k24-32 •
- ic:curve25519-family-k24-32 •
- ic:curve25519-family-k24-32 •
- ic:curve25519-family-k24-32 •
- ic:curve25519-family-k24-32 •
- ic:curve25519-family-k24 • s_w
- ic:curve25519-family-k28-32 •
- ic:curve25519-family-k28-32 •
- ic:curve25519-family-k28-32 •
- ic:curve25519-family-k28 • s_w
- ic:curve25519-family-k32 • s_w
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- rho:prime_medium_tier_bound •
- rho:prime_medium_tier_bound •
- rho:prime_medium_tier_bound •
- rho:prime_medium_tier_bound •
- rho:prime_medium_tier_bound •
- rho:prime_medium_tier_bound •
- rho:prime_medium_tier_bound •
- rho:prime_medium_tier_bound •
- rho:prime_medium_tier_bound •
- rho:prime_medium_tier_bound •
- rho:prime_medium_tier_bound •
- rho:prime_medium_tier_bound •
- rho:prime_medium_tier_bound •

## Distance remaining: value / reference (1.0 = parity; derived)




### Series key

- archive:ic-packed-filter-p32-2
- archive:ic-packed-filter-p32-2
- dlpwai:static-dh-p224-shape •
- dlpwai:static-dh-p224-shape •
- dlpwai:static-dh-p224-shape •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20 • s_w
- ic:curve25519-family-k24-28 •
- ic:curve25519-family-k24-28 •
- ic:curve25519-family-k24-32 •
- ic:curve25519-family-k24-32 •
- ic:curve25519-family-k24-32 •
- ic:curve25519-family-k24 • s_w
- ic:curve25519-family-k28-32 •
- ic:curve25519-family-k28-32 •
- ic:curve25519-family-k28-32 •
- ic:curve25519-family-k28 • s_w
- ic:curve25519-family-k32 • s_w
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe

## Series in unit count: gain vs first record (derived, %)

### Series key

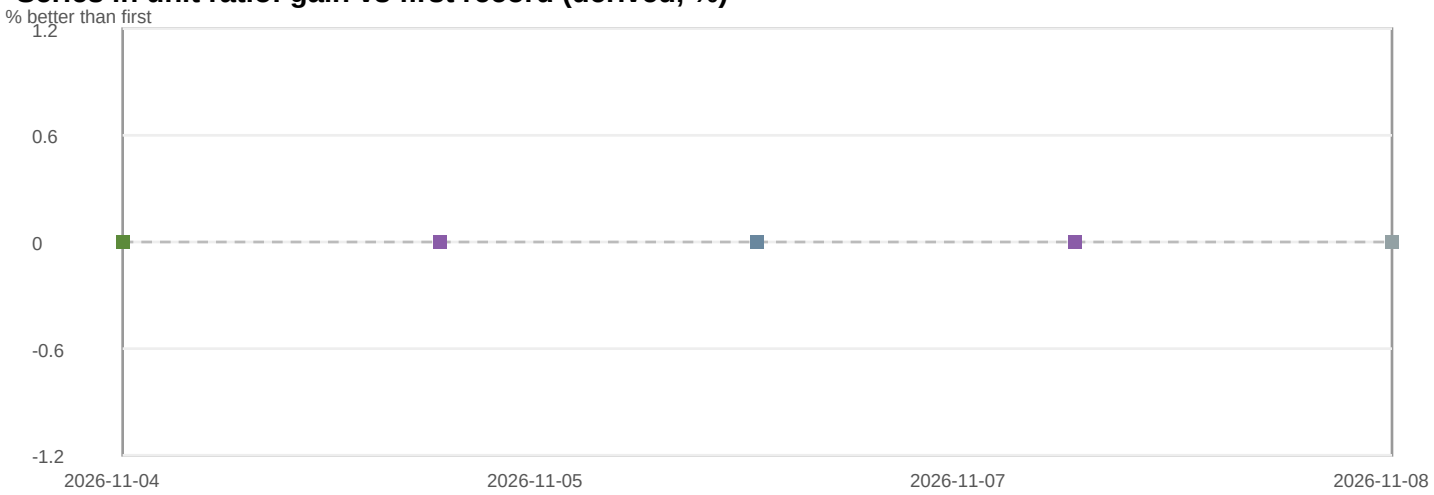
- analysis:ecbench_statistics_v2
- archive:icb-large-prime-n53-s1
- archive:icb-large-prime-n53-s2
- archive:icb-large-prime-n61-s1
- archive:icb-large-prime-n61-s2
- ecdlp:split-prime-p3-cross-tra
- ic:split-prime-p1-rr-dickson-t
- ic:split-prime-p2-twist-coset-
- registry-rebinding:ic-aimed-ga
- study:elliptic_code_list_decod

## Series in unit percent: gain vs first record (derived, %)

### Series key

- ecdlp:split-prime-p3-cross-tra
- ic:split-prime-p2-twist-coset-
- ic:split-prime-p2-twist-coset-

## Series in unit ratio: gain vs first record (derived, %)




### Series key

- analysis:ecbench_statistics_v2
- analysis:ecbench_statistics_v2
- archive:ic-packed-filter-p32-2
- ecdlp:split-prime-p3-cross-tra
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k20-32 •
- ic:curve25519-family-k24-32 •
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k20-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
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- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- ic:curve25519-k24-single-targe
- ic:split-prime-p2-twist-coset-
- kernel:f6_ic_gate • f4_control
- kernel:f6_ic_gate • f6ic_cpu_o
- kernel:f6_ic_gate • f6ic_cpu_o
- kernel:f6_ic_gate • f6ic_word_
- kernel:inherited_f4_solve_tree
- kernel:inherited_f4_solve_tree
- kernel:inherited_f4_solve_tree
- kernel:inherited_f4_solve_tree
- rho:koblitz_signed_frobenius_p
- rho:koblitz_signed_frobenius_p
- rho:koblitz_signed_frobenius_p
- rho:koblitz_signed_frobenius_p
- rho:koblitz_signed_frobenius_p

## Series summary

| problem • metric | unit | first | latest | gain vs first | value/ref | pts |
|---|---|---|---|---|---|---|
| analysis:ecbench_statistics_v2 • censored_ratio_ | ratio | 648.2 | 648.2 | - | - | 1 |
| analysis:ecbench_statistics_v2 • heavy_tail_rows | count | 71 | 71 | - | - | 1 |
| analysis:ecbench_statistics_v2 • ops_interval_wi | ratio | 13.1 | 13.1 | - | - | 1 |
| archive:ic-packed-filter-p32-20261029 • s_prime_ | S | 31.6936 | 31.6936 | 0% | 31.4718 | 1 |
| archive:ic-packed-filter-p32-20261029 • s_prime_ | S | 63.4961 | 63.4961 | 0% | 31.6255 | 1 |
| archive:ic-packed-filter-p32-20261029 • wall_per | ratio | 0.9601 | 0.9601 | 0% | - | 1 |
| archive:icb-large-prime-n53-s1-20261029 • report | count | 240 | 240 | - | - | 1 |
| archive:icb-large-prime-n53-s2-20261029 • report | count | 240 | 240 | - | - | 1 |
| archive:icb-large-prime-n61-s1-20261029 • report | count | 72 | 72 | - | - | 1 |
| archive:icb-large-prime-n61-s2-20261029 • report | count | 60 | 60 | - | - | 1 |
| dlpwai:static-dh-p224-shape • s_w0_cheon_pminus1 | S | 0.5739 | 0.5739 | 0% | 0.5814 | 2 |
| dlpwai:static-dh-p224-shape • s_w0_cheon_pminus1 | S | 12.2426 | 12.2426 | 0% | 11.6066 | 2 |
| dlpwai:static-dh-p224-shape • s_w0_cheon_pplus1_ | S | 0.5819 | 0.5819 | 0% | 0.5746 | 2 |
| ecdlp:split-prime-p3-cross-trace-census • aa_rho | ratio | 0.986 | 0.986 | 0% | - | 1 |
| ecdlp:split-prime-p3-cross-trace-census • bonfer | count | 0 | 0 | - | - | 1 |
| ecdlp:split-prime-p3-cross-trace-census • max_ab | percent | 0.7 | 0.7 | 0% | - | 1 |
| ic:curve25519-family-k20-32 • ratio_w0_k24 | ratio | 0.9375 | 0.9375 | 0% | - | 1 |
| ic:curve25519-family-k20-32 • ratio_w0_k32 | ratio | 0.8802 | 0.8802 | 0% | - | 1 |
| ic:curve25519-family-k20-32 • s_w0_incumbent_k20 | S | 2.5746 | 2.5746 | 0% | 1.7601 | 1 |
| ic:curve25519-family-k20-32 • s_w0_incumbent_k24 | S | 2.1638 | 1.6735 | 22.6592% | 1.429 | 2 |
| ic:curve25519-family-k20-32 • s_w0_incumbent_k28 | S | 1.9211 | 1.5717 | 18.1849% | 1.4943 | 2 |
| ic:curve25519-family-k20-32 • s_w0_incumbent_k32 | S | 2.11 | 1.5152 | 28.1881% | 1.5539 | 2 |
| ic:curve25519-family-k20-32 • s_w0_mean5curves_k | S | 2.601 | 2.601 | 0% | 1.5669 | 1 |
| ic:curve25519-family-k20-32 • s_w0_mean5curves_k | S | 2.111 | 2.111 | 0% | 1.7695 | 1 |
| ic:curve25519-family-k20-32 • s_w0_mean5curves_k | S | 1.869 | 1.869 | 0% | 1.7971 | 1 |
| ic:curve25519-family-k20-32 • s_w0_mean5curves_k | S | 1.93 | 1.93 | 0% | 1.9052 | 1 |
| ic:curve25519-family-k20 • s_w0_incumbent | S | 8.3478 | 4.8598 | 41.7827% | 3.2356 | 5 |
| ic:curve25519-family-k24-28 • s_w0_incumbent_k24 | S | 2.5842 | 2.5842 | 0% | 2.0304 | 1 |
| ic:curve25519-family-k24-28 • s_w0_incumbent_k28 | S | 2.3732 | 2.3732 | 0% | 2.3865 | 1 |
| ic:curve25519-family-k24-32 • best_train_s_w0_k2 | S | 2.713 | 2.713 | 0% | - | 1 |
| ic:curve25519-family-k24-32 • best_train_s_w0_k3 | S | 2.768 | 2.768 | 0% | - | 1 |
| ic:curve25519-family-k24-32 • ratio_w0_k32 | ratio | 0.9106 | 0.9106 | 0% | - | 1 |
| ic:curve25519-family-k24-32 • s_w0_seedrun_k24 | S | 2.1055 | 2.1055 | 0% | 1.942 | 1 |
| ic:curve25519-family-k24-32 • s_w0_seedrun_k28 | S | 1.8835 | 1.8835 | 0% | 1.8154 | 1 |
| ic:curve25519-family-k24-32 • s_w0_seedrun_k32 | S | 1.8423 | 1.8423 | 0% | 1.8232 | 1 |
| ic:curve25519-family-k24 • s_w0_incumbent | S | 6.607 | 3.3182 | 49.7782% | 2.7163 | 5 |
| ic:curve25519-family-k28-32 • s_w0_incumbent | S | 2.5798 | 2.5798 | 0% | 2.5891 | 1 |
| ic:curve25519-family-k28-32 • s_w0_seed_mean_k28 | S | 1.969 | 1.969 | 0% | 1.8593 | 1 |
| ic:curve25519-family-k28-32 • s_w0_seed_mean_k32 | S | 1.781 | 1.781 | 0% | 1.8229 | 1 |
| ic:curve25519-family-k28 • s_w0_incumbent | S | 10.9395 | 2.576 | 76.452% | 2.7572 | 7 |
| ic:curve25519-family-k32 • s_w0_incumbent | S | 24.1377 | 2.519 | 89.5641% | 2.7453 | 5 |
| ic:curve25519-k20-single-target • ratio_f2_over_ | ratio | 1.2283 | 1.2283 | -0% | - | 1 |
| ic:curve25519-k20-single-target • ratio_f2_over_ | ratio | 1.2385 | 1.2385 | -0% | - | 1 |
| ic:curve25519-k20-single-target • ratio_f2_over_ | ratio | 1.1542 | 1.1542 | -0% | - | 1 |
| ic:curve25519-k20-single-target • ratio_tw_over_ | ratio | 1.1656 | 1.1656 | 0% | - | 1 |
| ic:curve25519-k20-single-target • ratio_twcj_ove | ratio | 0.8872 | 0.8872 | 0% | - | 1 |
| ic:curve25519-k20-single-target • ratio_twcj_ove | ratio | 0.9088 | 0.9088 | 0% | - | 1 |
| ic:curve25519-k20-single-target • ratio_twcj_ove | ratio | 0.7335 | 0.7335 | 0% | - | 1 |
| ic:curve25519-k20-single-target • ratio_twcj_ove | ratio | 1.3058 | 1.3058 | 0% | - | 1 |
| ic:curve25519-k20-single-target • ratio_twcj_ove | ratio | 1.3963 | 1.3963 | 0% | - | 1 |
| ic:curve25519-k20-single-target • ratio_twcj_ove | ratio | 1.3602 | 1.3602 | 0% | - | 1 |
| ic:curve25519-k20-single-target • ratio_twcj_ove | ratio | 0.8312 | 0.8312 | 0% | - | 1 |
| ic:curve25519-k20-single-target • ratio_twcj_ove | ratio | 0.7797 | 0.7797 | 0% | - | 1 |
| ic:curve25519-k20-single-target • s_f1_x07_w01 | S | 2.3781 | 2.3781 | 0% | 1.4287 | 1 |
| ic:curve25519-k20-single-target • s_f1_x07_w0 | S | 2.3211 | 2.3211 | 0% | 1.4951 | 1 |

| | | | | | | |
|---|---|---|---|---|---|---|
| ic:curve25519-k20-single-target • s_f1_x07_w1 | S | 2.8907 | 2.8907 | 0% | 1.0818 | 1 |
| ic:curve25519-k20-single-target • s_twcj_w01 | S | 2.1428 | 2.1428 | 0% | 1.3059 | 1 |
| ic:curve25519-k20-single-target • s_twcj_w0 | S | 2.1383 | 2.1383 | 0% | 1.3963 | 1 |
| ic:curve25519-k20-single-target • s_twcj_w1 | S | 2.183 | 2.183 | 0% | 0.8311 | 1 |
| ic:curve25519-k20-single-target • twcj_walk_s | S | 1.3895 | 1.3895 | 0% | - | 1 |
| ic:curve25519-k24-single-target • ratio_f2_over_ | ratio | 1.1475 | 1.1475 | -0% | - | 1 |
| ic:curve25519-k24-single-target • ratio_f2_over_ | ratio | 1.1636 | 1.1636 | -0% | - | 1 |
| ic:curve25519-k24-single-target • ratio_f2_over_ | ratio | 1.0384 | 1.0384 | -0% | - | 1 |
| ic:curve25519-k24-single-target • ratio_tw_over_ | ratio | 0.9801 | 0.9801 | 0% | - | 1 |
| ic:curve25519-k24-single-target • ratio_twcj_ove | ratio | 0.8609 | 0.8609 | 0% | - | 1 |
| ic:curve25519-k24-single-target • ratio_twcj_ove | ratio | 0.8882 | 0.8882 | 0% | - | 1 |
| ic:curve25519-k24-single-target • ratio_twcj_ove | ratio | 0.675 | 0.675 | 0% | - | 1 |
| ic:curve25519-k24-single-target • ratio_twcj_ove | ratio | 1.2961 | 1.2961 | 0% | - | 1 |
| ic:curve25519-k24-single-target • ratio_twcj_ove | ratio | 1.4114 | 1.4114 | 0% | - | 1 |
| ic:curve25519-k24-single-target • ratio_twcj_ove | ratio | 1.3883 | 1.3883 | 0% | - | 1 |
| ic:curve25519-k24-single-target • ratio_twcj_ove | ratio | 0.7492 | 0.7492 | 0% | - | 1 |
| ic:curve25519-k24-single-target • ratio_twcj_ove | ratio | 0.9063 | 0.9063 | 0% | - | 1 |
| ic:curve25519-k24-single-target • s_f1_x07_w01 | S | 1.8945 | 1.8945 | 0% | 1.4975 | 1 |
| ic:curve25519-k24-single-target • s_f1_x07_w0 | S | 1.8341 | 1.8341 | 0% | 1.5802 | 1 |
| ic:curve25519-k24-single-target • s_f1_x07_w1 | S | 2.4383 | 2.4383 | 0% | 1.1058 | 1 |
| ic:curve25519-k24-single-target • s_twcj_w01 | S | 1.6241 | 1.6241 | 0% | 1.2961 | 1 |
| ic:curve25519-k24-single-target • s_twcj_w0 | S | 1.623 | 1.623 | 0% | 1.4113 | 1 |
| ic:curve25519-k24-single-target • s_twcj_w1 | S | 1.6342 | 1.6342 | 0% | 0.7492 | 1 |
| ic:curve25519-k24-single-target • twcj_walk_s | S | 1.394 | 1.394 | 0% | - | 1 |
| ic:split-prime-p1-rr-dickson-torus • capped_jobs | count | 8 | 8 | 0% | - | 1 |
| ic:split-prime-p2-twist-coset-toy • coverage_law | percent | 0.93 | 0.93 | 0% | - | 1 |
| ic:split-prime-p2-twist-coset-toy • f4_work_rel_ | percent | 0.39 | 0.39 | 0% | - | 1 |
| ic:split-prime-p2-twist-coset-toy • gain_min_t4 | ratio | 2.25 | 2.25 | -0% | - | 1 |
| ic:split-prime-p2-twist-coset-toy • purity_nontr | count | 0 | 0 | - | - | 1 |
| kernel:f6_ic_gate • f4_control_word_xors_off_ove | ratio | 1 | 1 | - | - | 1 |
| kernel:f6_ic_gate • f6ic_cpu_off_over_on_max | ratio | 10.29 | 10.29 | -0% | - | 1 |
| kernel:f6_ic_gate • f6ic_cpu_off_over_on_min | ratio | 2.67 | 2.67 | -0% | - | 1 |
| kernel:f6_ic_gate • f6ic_word_xors_off_over_on_m | ratio | 1.84 | 1.84 | -0% | - | 1 |
| kernel:inherited_f4_solve_tree • wall_seq_over_p | ratio | 0.936 | 0.936 | -0% | - | 1 |
| kernel:inherited_f4_solve_tree • wall_seq_over_p | ratio | 2.883 | 2.883 | -0% | - | 1 |
| kernel:inherited_f4_solve_tree • wall_seq_over_p | ratio | 1.589 | 1.589 | -0% | - | 1 |
| kernel:inherited_f4_solve_tree • work_par_over_s | ratio | 2.865 | 2.865 | 0% | - | 1 |
| registry-rebinding:ic-aimed-gather • raw_pins_re | count | 211 | 211 | - | - | 1 |
| rho:koblitz_signed_frobenius_panel • s_legacy_ov | ratio | 0.718 | 0.718 | -0% | - | 1 |
| rho:koblitz_signed_frobenius_panel • s_legacy_ov | ratio | 1.0948 | 1.0948 | -0% | - | 1 |
| rho:koblitz_signed_frobenius_panel • s_legacy_ov | ratio | 1.4723 | 1.4723 | -0% | - | 1 |
| rho:koblitz_signed_frobenius_panel • s_legacy_ov | ratio | 1.0653 | 1.0653 | -0% | - | 1 |
| rho:koblitz_signed_frobenius_panel • s_legacy_ov | ratio | 1.0008 | 1.0008 | -0% | - | 1 |
| rho:prime_medium_tier_bound • prime-medium/bsgs- | S | 1.0827 | 1.0827 | 0% | - | 1 |
| rho:prime_medium_tier_bound • prime-medium/bsgs- | S | 0.9842 | 0.9842 | 0% | - | 1 |
| rho:prime_medium_tier_bound • prime-medium/bsgs- | S | 1.9684 | 1.9684 | 0% | - | 1 |
| rho:prime_medium_tier_bound • prime-medium/rho-n | S | 1.0422 | 1.0422 | 0% | - | 1 |
| rho:prime_medium_tier_bound • prime-medium/rho-n | S | 0.9473 | 0.9473 | 0% | - | 1 |
| rho:prime_medium_tier_bound • prime-medium/rho-n | S | 1.8961 | 1.8961 | 0% | - | 1 |
| rho:prime_medium_tier_bound • prime-medium/rho-n | S | 1.0658 | 1.0658 | 0% | - | 1 |
| rho:prime_medium_tier_bound • prime-medium/rho-n | S | 0.969 | 0.969 | 0% | - | 1 |
| rho:prime_medium_tier_bound • prime-medium/rho-n | S | 1.9367 | 1.9367 | 0% | - | 1 |
| rho:prime_medium_tier_bound • prime-medium/rho-p | S | 1.2703 | 1.2703 | 0% | - | 1 |
| rho:prime_medium_tier_bound • prime-medium/rho-p | S | 1.2703 | 1.2703 | 0% | - | 1 |
| rho:prime_medium_tier_bound • prime-medium/rho-p | S | 1.2703 | 1.2703 | 0% | - | 1 |
| study:elliptic_code_list_decoding_20261107 • d1_ | count | 0 | 0 | - | - | 1 |

Every value is quoted from the record it names, in that record’s unit. Percentages and ratios under ’derived’ are computed here from those values and are not measurements. Series in different units are never compared.